

Lukas R. Stone

PhD Candidate
The Pennsylvania State University

Summary

Research: High-Energy instrumentation, X-ray detector development, X-ray hybrid CMOS detector characterization and optimization, CubeSAT development and operation, Test stand design, Jetted tidal disruption events, High-redshift Gamma-ray bursts

Techniques: X-ray imaging, Non-dispersive X-ray Spectroscopy, X-ray data analysis, z-transform auto-discrete correlation analysis, CubeSat assembly and testing

Publications: 7 as lead/co-lead author, 22 total, 83 citations, 2 h-index

Presentations: 9 research, 9 outreach and professional development

Academic and Professional Appointments

2021- Graduate Student (Ph.D.) – The Pennsylvania State University

2019-21 Research Assistant (B.S.) – University of Oklahoma

Education

Ph.D. Astronomy & Astrophysics | *Cumulative GPA: 3.97/4.0* The Pennsylvania State University 2021 -

Advisor: Dr. Abraham D. Falcone

Thesis: *Characterization of Low-noise Fine-resolution X-ray Hybrid CMOS Detectors*

B.S. Astrophysics | *Cumulative GPA: 4.0/4.0* University of Oklahoma 2016 - 2020

Advisor: Dr. Nathan A. Kaib

Undergraduate Research: *Evolution of primordial Kuiper Belt binaries through a giant planet instability*

Awarded Proposals, Grants & Fellowships

2026 Larry Travis Graduate Student Research Fellowship \$400

Honors and Awards

2021 Homer F. Braddock Scholarship in Biology, Chemistry, and Physics OU

2020 4.0 Medallion for Academic Excellence in Achieving an Overall 4.0 Grade Point Average at the University of Oklahoma OU

2019 Duane E. Roller Award for Outstanding Scholarship in Physics and Astronomy OU

2017-20 Homer L. Dodge Department of Physics and Astronomy Meritorious Scholarship OU

Teaching Experience

Teaching Assistant/Course Assistant

Astro 320W: Observational Astronomy Laboratory

Penn State

Service

Department

Astronomy and Astrophysics Graduate Student Representative	Penn State
Member – Climate & Diversity Committee	Penn State
Member – Facilities and Safety/Office Spaces Committee	Penn State
Member – Prospective Graduate Student Visit Planning Committee	Penn State
Member – Recruitment Conferences Committee	Penn State

Mentoring

Students Supervised

Primary mentor for 2 undergraduate researchers in the High-Energy Astrophysical Detector & Instrumentation Lab at Penn State on student-led research projects	2024-
Co-mentor for 9 undergraduate researchers in the High-Energy Astrophysical Detector & Instrumentation Lab at Penn State	2021-2026

Other Mentoring Activities

Coordinator – Penn State Graduate Student Mentoring Program	Penn State
Mentor – Penn State Graduate Student Mentoring Program	Penn State

Outreach

Presenter – Central Pennsylvania Observers	Penn State
Presenter – Astronomy on Tap State College	Penn State
Volunteer – AstroFest	Penn State

Technical Skills and Programming Languages

Languages	Fluent: Python, LaTeX Familiar: LabVIEW, Julia, C, Fortran, HTML, bash
Software	Fluent: pyzdcf, DS9, xspec, ximage, xrtgrblc Familiar: IRAF, AstrolmageJ, sbatch, slurm
Hardware	Testbed design and development, Detector development, CubeSat integration and assembly, Vibrational Testing, Thermal Vacuum testing

Website Architect

Personal Website: LukasStone.github.io
 High-Energy Astrophysical Detector & Instrumentation lab: sites.psu.edu/headilab

Societies and Organizations

2025	American Astronomical Society	Member
2022	Society of Photographic Instrumentation Engineers	Member

Presentations

Research Presentations (**bold** = invited)

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| 9. | Talk: <i>The BlackCAT CubeSat: a soft X-ray coded aperture telescope</i> at SPIE: Astronomical Telescopes & Instrumentation 2026, Copenhagen, Denmark | 9 July 2026 |
| 8. | Talk: <i>A Low-Noise Small-pixel Hybrid CMOS Detector for future X-ray Observatories</i> at SPIE: Astronomical Telescopes & Instrumentation 2026, Copenhagen, Denmark | 5 July 2026 |
| 7. | Talk: <i>Fine-Resolution Low-Noise X-ray Hybrid CMOS Detectors For Future High-Throughput X-ray Missions</i> at AAS 22 nd HEAD Meeting, St. Louis, Missouri, USA | 15 October 2025 |
| 6. | Poster: <i>Status of Testing and Characterization of a Low-noise Small-Pixel Hybrid CMOS Detector</i> at SPIE: Optics + Photonics 2025, San Diego, California, USA | 4 August 2025 |
| 5. | Talk: <i>Status of X-ray Hybrid CMOS Detector Development Efforts at PSU</i> at SPIE: Astronomical Telescopes & Instrumentation 2024, Yokohama, Japan | 17 June 2024 |
| 4. | Talk: <i>Status of Testing and Characterization of the Speedster-EXD550 X-ray Hybrid CMOS Detector</i> at SPIE: Optics + Photonics 2023, San Diego, California, USA | 21 August 2023 |
| 3. | Talk: <i>X-ray Hybrid CMOS Detectors: Status of the Event-Driven Speedster-EXD550 & Introducing the Next-gen Small-Pixel Detector</i> , The Pennsylvania State University, State College, Pennsylvania, USA | 7 November 2023 |
| 2. | Talk: <i>Gain Variability in a Speedster-EXD550 Hybrid CMOS X-ray Detector</i> , The Pennsylvania State University, State College, Pennsylvania, USA | 31 January 2023 |
| 1. | Talk: <i>Astrophysical Detectors and their application to X-ray astronomy</i> , The Pennsylvania State University, State College, Pennsylvania, USA | 15 March 2022 |

Outreach Presentations

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| 5. | Talk: <i>Science in a shoebox: The BlackCAT CubeSat</i> for the Central Pennsylvania Observers, State College, Pennsylvania, USA | 1 October 2026 |
| 4. | Talk: <i>Science in a shoebox: The BlackCAT CubeSat</i> at Astronomy on Tap State College, State College, Pennsylvania, USA | 22 April 2026 |
| 3. | Talk: <i>Cosmic Connections: Astronomy, Spirit, and Our Shared Responsibility</i> , State College, Pennsylvania, USA | 24 August 2025 |
| 2. | Talk: <i>Red Giant Rising: When Your Landlord Becomes a Fireball: Eviction notice: 5 billion years</i> at Astronomy on Tap State College, State College, Pennsylvania, USA | 22 May 2025 |
| 1. | Talk: <i>X-rays: Röntgen to Vision</i> at Astronomy on Tap State College, State College, Pennsylvania, USA | 1 March 2023 |

Professional Development Presentations

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| 4. | Panel: <i>Surviving your First Year in Graduate School</i> at The Pennsylvania State University, State College, Pennsylvania, USA | 20 August 2025/26 |
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| 3. | Talk: <i>Applying to Graduate School Panel</i> at The Pennsylvania State University, State College, Pennsylvania, USA | 1 October 2024 |
| 2. | Talk: <i>C&D Astro+ Research Showcase</i> at The Pennsylvania State University, State College, Pennsylvania, USA | 8 November 2024 |
| 1. | Talk: <i>Applying to Graduate School Panel</i> at The Pennsylvania State University, State College, Pennsylvania, USA | 1 October 2024 |

Publications

Lead/Co-Lead

7. **Stone, L.R.**, Falcone, A.D., and Griffith, C.V., 2026. "*Revisiting X-ray Periodicities in the Transient Source SWIFT J1644+57*." arXiv e-prints, arXiv:XYZ (Submitted to ApJ)
6. **Stone, L.R.**, Falcone, A.D., et al., 2026. "*Low-Noise Small-pixel X-ray Hybrid CMOS Detectors for Next-generation X-ray Missions*." arXiv e-prints, arXiv:XYZ (Submitted to JATIS)
5. **Stone, L.R.**, Falcone, A.D., et al., 2026. "*A low-noise small-pixel hybrid CMOS detector for future X-ray observatories*." In *X-Ray, Optical, and Infrared Detectors for Astronomy XII* (Vol. 14157, pp. 164-169). SPIE.
4. **Stone, L.R.**, Falcone, A.D., et al., 2025. "*Status of testing and characterization of a low-noise small-pixel hybrid CMOS detector*." In *UV, X-Ray, and Gamma-Ray Space Instrumentation for Astronomy XXIV* (Vol. 13625, pp. 581-586). SPIE
3. **Stone, L.R.**, Falcone, A.D., et al., 2024. "*Status of X-ray hybrid CMOS detector development efforts at PSU*." In *X-Ray, Optical, and Infrared Detectors for Astronomy XI* (Vol. 13103, pp. 235-246). SPIE.
2. **Stone, L.R.**, Falcone, A.D., et al., 2023. "*Status of testing and characterization of the Speedster-EXD550 x-ray hybrid CMOS detector*." In *UV, X-Ray, and Gamma-Ray Space Instrumentation for Astronomy XXIII* (Vol. 12678, pp. 218-222). SPIE.
1. **Stone, L.R.** and Kaib, N.A., 2021. "*Evolution of primordial Kuiper belt binaries through a giant planet instability*." *Monthly Notices of the Royal Astronomical Society: Letters*, 505(1), pp.L31-L35.

Co-Author: Published

15. Falcone, A.D., ... **Stone, L.R.**, et al., 2026, "*The BlackCAT CubeSat: a soft x-ray coded aperture telescope*." In *Space Telescopes and Instrumentation 2026: Ultraviolet to Gamma Ray* (p. PC1414610). SPIE.
14. O'Neill, L., ... **Stone, L.R.**, et al., 2026, "*Source centroiding and angular resolution with BlackCAT and beyond*." In *X-Ray, Optical, and Infrared Detectors for Astronomy XII* (Vol. 14157, pp. 749-756). SPIE.
13. Emeigh, T.R., ... **Stone, L.R.**, et al., 2026, "*Development of next-generation event-driven x-ray hybrid CMOS detectors*." In *X-Ray, Optical, and Infrared Detectors for Astronomy XII* (Vol. 14157, pp. 140-147). SPIE.
12. Hossen, Md. A., ... **Stone, L.R.**, et al., 2026, "*Quantum efficiency of the Speedster-EXD550 hybrid CMOS detector in the soft X-ray band*." *Journal of Astronomical Telescopes, Instruments, and Systems*, 12(3), pp.036002-036002.
11. Bevidas Jr, W.A., ... **Stone, L.R.**, et al., 2025, "*Characterization of random telegraph noise in an H2RG X-ray hybrid CMOS detector*." *Journal of Astronomical Telescopes, Instruments, and Systems*, 11(2), p.026002.
10. Emeigh, T.R., ... **Stone, L.R.**, et al., 2025, "*Calibration measurements for the BlackCAT CubeSat x-ray coded aperture telescope*." In *UV, X-Ray, and Gamma-Ray Space Instrumentation for Astronomy XXIV* (Vol. 13625, pp. 517-524). SPIE.
9. Wages, M., ... **Stone, L.R.**, et al., 2025, "*Mechanical design and implementation of the BlackCAT payload*." In *UV, X-Ray, and Gamma-Ray Space Instrumentation for Astronomy XXIV* (Vol. 13625, pp. 322-335). SPIE.

8. Falcone, A. D., ... **Stone, L.R.**, et al., 2024, "*BlackCAT: an upcoming soft x-ray coded aperture telescope on a 6U CubeSat.*" In Space Telescopes and Instrumentation 2024: Ultraviolet to Gamma Ray (Vol. 13093, pp. 334-347). SPIE.
7. Colosimo, J. M., ... **Stone, L.R.**, et al., 2024, "*Ground calibration of the BlackCAT CubeSat x-ray Coded Aperture Telescope.*" In Space Telescopes and Instrumentation 2024: Ultraviolet to Gamma Ray (Vol. 13093, pp. 2027-2034). SPIE.
6. Colosimo, Joseph M., ... **Stone, L.R.**, et al., 2024, "*Expected Gamma-Ray Burst Detection Rates and Redshift Distributions for the BlackCAT CubeSat Mission.*" The Astrophysical Journal, 969(2), p.138.
5. Colosimo, J., ...**Stone, L.R.**, et al., 2023, "*Current status of the BlackCAT CubeSat.*" In UV, X-Ray, and Gamma-Ray Space Instrumentation for Astronomy XXIII (Vol. 12678, pp. 288-290). SPIE.
4. Colosimo, J., Grzybowski, H., Jennerjahn, E., **Stone, L.R.**, Falcone, A.D., Wages, M., et al., 2023, "*Initial Characterization of the first Speedster-EXD550 Event-Driven X-ray Hybrid CMOS Detectors,*" Journal of Astronomical Telescopes, Instruments, and Systems, 9(4), pp.046002-046002.
3. Campbell, H., **Stone, L.R.**, and Kaib, N.A., 2023, "*Close Trans-Neptunian Object Passages as a Driver of the Origin and Evolution of Ultrawide Kuiper Belt Binaries.*" The Astronomical Journal, 165(1), p.19.
2. Colosimo, J., ...**Stone, L.R.**, et al., 2022, "*Estimating the background of the BlackCAT CubeSat and its impact on science observations.*" In Space Telescopes and Instrumentation 2022: Ultraviolet to Gamma Ray (Vol. 12181, pp. 1322-1331). SPIE.
1. Baker, L., ...**Stone, L.R.**, et al., 2022, "*Mechanical and thermal design of the BlackCAT CubeSat.*" In Space Telescopes and Instrumentation 2022: Ultraviolet to Gamma Ray (Vol. 12181, pp. 1310-1321). SPIE.

Co-Author: Submitted

1. Emeigh, T.R., ... **Stone, L.R.**, et al., 2025, "*Calibration measurements for the BlackCAT CubeSat x-ray coded aperture telescope.*" arXiv e-prints, arXiv:XYZ (Submitted to JATIS)

References

Ph.D. Advisor

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